Mathematical Models in Biotechnology

Jeffrey D. Varner
Robert Frederick Smith School of Chemical and Biomolecular Engineering,
Cornell University, Ithaca, NY, USA

Abstract

Intracellular fluxes, kinetic parameters, and metabolite concentrations are sparsely measured at genome scale. Flux balance analysis estimates fluxes from network stoichiometry, reaction bounds, and an optimization objective. This chapter derives the flux constraint from a mole balance and states the assumptions that lead to its steady-state form. It then writes each reaction bound in terms of reversibility, enzyme capacity, substrate saturation, enzyme abundance, and regulation. A urea-cycle model shows how public thermodynamic and kinetic estimates set reaction bounds. A feedback-inhibited pathway shows how expression and allosteric regulation change a capacity bound. Two larger studies illustrate related methods in cell-free protein synthesis and strain design. Together, these examples show how condition-specific data and proposed interventions change flux predictions through reaction bounds.

1 Introduction

Biotechnology depends on predicting and directing cellular metabolism. Metabolic engineering redirects flux toward a product. Bioprocess development selects operating conditions that sustain production. Strain design identifies genetic changes that alter the phenotype. Each task requires an estimate of metabolic reaction rates under a specified condition [1]. These rates are difficult to measure directly. Genome sequences and reaction annotations provide a network structure, but reconstructions remain uncertain in reaction membership, directionality, compartments, cofactors, and biomass composition. Fluxes, kinetic parameters, and metabolite concentrations are even less complete at genome scale. Models must therefore estimate phenotype from incomplete measurements. Bioprocess models use several levels of biological detail. Unstructured models track biomass, extracellular substrates, and products with lumped growth, uptake, and production rates. Monod growth, inhibition, maintenance, and growth-associated production laws support batch, fed-batch, and continuous reactor analysis, but do not resolve intracellular fluxes [2]. Structured kinetic models add intracellular pools, enzymes, or cellular machinery. Early single-cell models from Shuler and coworkers coupled intracellular metabolism to extracellular nutrients in microbial and mammalian cells [3, 4]. Reaction-level kinetic models assign mass-action, Michaelis–Menten, or power-law rates to individual reactions and integrate the resulting differential equations [5–8]. These models describe transient intracellular behavior, but require more kinetic parameters as their resolution increases.

Constraint-based models use a different closure. They impose mass conservation on the reaction network and require fewer explicit kinetic parameters. This approach can be applied to genome-scale reconstructions [1, 9]. Flux balance analysis (FBA) formulates the calculation as a linear program [10, 11]. Conservation alone usually permits many flux distributions. Reaction bounds restrict this feasible set, and an objective selects an optimum. Bounds are a major route by which condition-specific thermodynamic, kinetic, expression, and regulatory data enter the model. This chapter examines how each type of data changes a bound and its predicted fluxes. The chapter first derives the flux constraint from a mole balance, including the growth-dilution balance $\mathbf{S}\hat{\mathbf{v}} = \mu\mathbf{x}$, where $\mathbf{S}$ is the stoichiometric matrix, $\hat{\mathbf{v}}$ is the intracellular flux vector, μ is the growth rate, and $\mathbf{x}$ is the intracellular concentration vector. It then factors each reaction bound into terms for reversibility, enzyme capacity, substrate saturation, enzyme abundance, and regulation. Two small examples show how these terms affect a flux prediction. The first uses public data to set bounds in a urea-cycle network. The second transfers expression and metabolite states from a kinetic feedback model into an FBA bound. Two larger studies illustrate related methods in dynamic cell-free protein synthesis [12, 13] and model-guided strain design [14]. Strain design also requires a discrete search over candidate interventions, often formulated as a bilevel optimization problem [15].

2 From Mole Balances to the Flux Constraint

The FBA constraint follows from a mole balance. Its derivation defines the normalization basis and identifies the steady-state assumptions. Consider a well-mixed culture with biomass concentration B and reactor volume $\bar{V}$. The biomass basis is therefore given by:

$$\mathcal{B} \equiv B\bar{V}, \tag{1}$$

where B has units gDW/L, $\bar{V}$ has units L, and $\mathcal{B}$ is the total biomass in gDW. An intracellular concentration C_i has units mmol/gDW, and an intracellular flux $\hat{v}_j$ has units mmol/gDW/h. Multiplication by the biomass basis $\mathcal{B}$ gives the total intracellular amount $C_i\mathcal{B}$ and total reaction rate $\hat{v}_j\mathcal{B}$. A general intracellular balance includes reaction, active transport, and physical transport. Let $c_{i,s}$ be the concentration of species i in a hypothetical bulk stream s, and let $\dot{V}_s$ be its volumetric flow rate. The intracellular mole balance is given by:

$$\sum_s d_s c_{i,s}\dot{V}_s + \mathcal{B}\sum_j \sigma_{ij}\hat{v}_j = \frac{d}{dt}\big(C_i\mathcal{B}\big), \tag{2}$$

where the stream-direction coefficient $d_s = +1$ for an inlet and $d_s = -1$ for an outlet, and σ_{ij} is entry (i, j) of the stoichiometric matrix $\mathbf{S} \in \mathbb{R}^{m\times n}$ for m metabolites and n reactions. Each term has units mmol/h. The stream concentration $c_{i,s}$ has units mmol/L, whereas C_i has units mmol/gDW. The cell membrane excludes bulk physical flow from the intracellular control volume, so the volumetric flow rate $\dot{V}_s = 0$ and the first term vanishes. Active transport remains in the flux vector $\hat{\mathbf{v}}$ because it moves molecules without bulk volume flow.

Expand the accumulation term before assuming a fixed reactor volume. Substitution of the biomass-basis relation $\mathcal{B} = B\bar{V}$ gives:

$$\frac{d}{dt}\big(C_i\mathcal{B}\big) = B\bar{V}\frac{dC_i}{dt} + C_i\bar{V}\frac{dB}{dt} + C_iB\frac{d\bar{V}}{dt}. \tag{3}$$

After division by the biomass basis $\mathcal{B} = B\bar{V}$, Equation (2) becomes:

$$\frac{1}{\mathcal{B}}\sum_s d_s c_{i,s}\dot{V}_s + \sum_j \sigma_{ij}\hat{v}_j = \frac{dC_i}{dt} + \frac{C_i}{B}\frac{dB}{dt} + \frac{C_i}{\bar{V}}\frac{d\bar{V}}{dt}. \tag{4}$$

For a batch or fed-batch culture with no biomass flow, total biomass changes through growth. The specific growth rate is given by:

$$\mu \equiv \frac{1}{\mathcal{B}}\frac{d\mathcal{B}}{dt} = \frac{1}{B}\frac{dB}{dt} + \frac{1}{\bar{V}}\frac{d\bar{V}}{dt}. \tag{5}$$

The membrane removes the physical-flow term from Equation (4). Define the concentration vector as $\mathbf{x} = (C_1, \ldots, C_m)^\top$ in mmol/gDW. Combining the remaining terms with Equation (5) gives the dynamic intracellular balance:

$$\mathbf{S}\hat{\mathbf{v}} = \frac{d\mathbf{x}}{dt} + \mu\mathbf{x}. \tag{6}$$

Balanced growth gives $d\mathbf{x}/dt = \mathbf{0}$, so this balance reduces to:

$$C_i\mu = \sum_j \sigma_{ij}\hat{v}_j \qquad \Longleftrightarrow \qquad \boxed{\mathbf{S}\hat{\mathbf{v}} = \mu\mathbf{x}}, \tag{7}$$

where the flux vector $\hat{\mathbf{v}} \in \mathbb{R}^n$ contains the reaction fluxes in mmol/gDW/h. Net production balances growth dilution. For a fixed-volume batch reactor, $d\bar{V}/dt = 0$ and $\mu = (1/B)dB/dt$. For fed-batch operation, $d\bar{V}/dt \neq 0$, so the volume term must remain. The fixed-volume result is a special case of the general balance.

Conventional FBA neglects the dilution term and uses the following approximation:

$$\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}. \tag{8}$$

This approximation is appropriate when the growth-dilution term $\mu\mathbf{x}$ is small relative to the metabolic fluxes [10]. Setting the growth rate $\mu = 0$ is not sufficient. A cell-free batch has no growing biomass, but its transcript, protein, and metabolite concentrations may still change. Its accumulation term $d\mathbf{x}/dt$ must remain; on a fixed concentration basis, Equation (6) becomes the dynamic balance $\mathbf{S}\mathbf{v} = d\mathbf{x}/dt$, where $\mathbf{v}$ is the cell-free flux vector. Dynamic constraint-based models of cell-free protein synthesis have used this form to impose time-dependent concentration changes over short intervals [12]. All model classes introduced above use the same balance but close the reaction term differently. Unstructured models use lumped rates, structured kinetic models use state-dependent intracellular rates, and constraint-based models use bounds and an objective. Constraint-based reconstructions represent uptake and secretion with transport and exchange reactions. A boundary reaction $\emptyset \rightleftharpoons M_i$ enters the stoichiometric matrix $\mathbf{S}$ as an isolated nonzero column with no intracellular reaction partner [1]. Its flux describes molecular transport across the modeled boundary, not a reactor stream. The included boundary reactions and their bounds define the modeled system boundary.

3 The Linear Program and Its Geometry

The balance equations conserve mass but usually admit many flux distributions; flux bounds restrict this set to a feasible polytope, and an objective selects an optimal vertex or face (Fig. 1). FBA adds lower and upper bounds, represented by the lower-bound vector $\boldsymbol{\ell}$ and upper-bound vector $\mathbf{u}$, and a linear objective with coefficient vector $\mathbf{c}$. The resulting constraints and objective are $\boldsymbol{\ell} \leq \hat{\mathbf{v}} \leq \mathbf{u}$ and $\mathbf{c}^\top\hat{\mathbf{v}}$, respectively. Bounds can represent reaction direction, enzyme capacity, and substrate availability. The objective states the quantity to maximize. Together, these elements define the linear program:

$$\begin{aligned} \max_{\hat{\mathbf{v}} \in \mathbb{R}^n} \quad & \mathbf{c}^\top\hat{\mathbf{v}} \\ \text{subject to} \quad & \mathbf{S}\hat{\mathbf{v}} = \mathbf{0} \ \ \text{(or } \mu\mathbf{x}\text{)}, \\ & \boldsymbol{\ell} \leq \hat{\mathbf{v}} \leq \mathbf{u}. \end{aligned} \tag{9}$$

This is the standard FBA formulation [10]. The objective is a modeling choice. Setting the objective coefficient $c_j = 1$ for a biomass pseudo-reaction maximizes growth. The biomass reaction consumes precursors in the proportions needed to form new cell mass. Setting the objective coefficient $c_j = 1$ for a product-export reaction instead maximizes product secretion. The constraints remain the same; only the objective coefficient vector $\mathbf{c}$ changes. The growth-aware form remains linear when the concentration vector $\mathbf{x}$ is measured and the growth rate μ is fixed or enters as a scalar decision variable. If both the growth rate μ and concentration vector $\mathbf{x}$ are unknown, their product is bilinear and the problem is no longer an LP. A model that retains the growth-dilution term $\mu\mathbf{x}$ must also define its relation to the biomass pseudo-reaction so that growth-associated demands are not counted twice. The conservation equations usually admit many flux vectors. The corresponding null space is given by:

$$\mathcal{N}(\mathbf{S}) = \{\hat{\mathbf{v}} \in \mathbb{R}^n : \mathbf{S}\hat{\mathbf{v}} = \mathbf{0}\}.$$

This set contains every steady-state flux vector. If the rank condition $\operatorname{rank}(\mathbf{S}) < n$ holds, the null space contains more than one vector. Bounds restrict it to a feasible polytope. The objective selects a vertex or a face of that polytope. A face contains multiple optimal flux vectors, so optimization does not always give a unique solution.

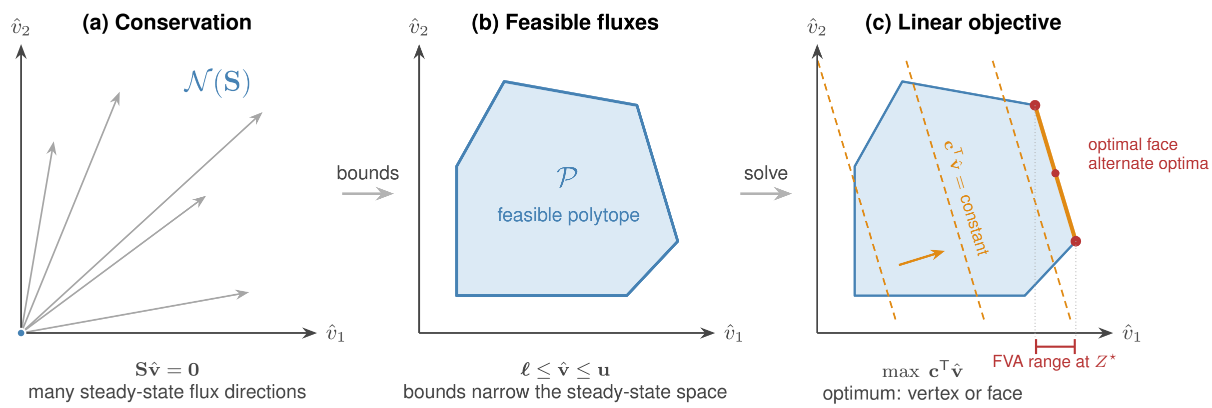

Figure 1: Geometry of FBA in a two-flux projection. (a) Conservation restricts steady-state fluxes to the null space $\mathcal{N}(\mathbf{S})$. (b) Lower and upper bounds form the feasible polytope $\mathcal{P}$. (c) The objective moves toward larger values until it reaches an optimal vertex or face. Fluxes can vary along an optimal face without changing the objective. FVA reports the resulting interval for each reaction, shown here for reaction flux $\hat{v}_1$.

The singular value decomposition identifies both null spaces. It factors the stoichiometric matrix as $\mathbf{S} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, where $\mathbf{U}$ is the left-singular-vector matrix, $\mathbf{\Sigma}$ is the diagonal singular-value matrix, and $\mathbf{V}$ is the right-singular-vector matrix. Columns of the right-singular-vector matrix $\mathbf{V}$ associated with zero singular values span the right null space, so every steady-state flux vector is a linear combination of these columns. Columns of the left-singular-vector matrix $\mathbf{U}$ associated with zero singular values span the left null space. A coefficient vector $\mathbf{y}$ in the left null space defines the conserved quantity $\mathbf{y}^\top\mathbf{x}$, such as the total concentration of a cofactor pair. Genome-scale models often have a large right null space because the rank condition $\operatorname{rank}(\mathbf{S}) < n$ holds [1]. Flux variability analysis (FVA) measures the remaining freedom in each reaction [11]. FVA fixes the objective at its optimum, or at a chosen fraction of that optimum, and then minimizes and maximizes each flux. Equal minimum and maximum values mean that a reaction is fixed on that optimal face. A wide interval means that the reaction can vary without changing the required objective value. A different objective can select a different face and produce different intervals.

4 Integrative Flux Bounds

Reaction bounds can include several types of condition-specific data. For reaction j, the bound is written as:

$$-\delta_j \left[V^{\circ}_{\max,j}\,(e/e^{\circ})\,\theta_j(\cdot)\,f_j(\cdot)\right] \;\leq\; \hat{v}_j \;\leq\; V^{\circ}_{\max,j}\,(e/e^{\circ})\,\theta_j(\cdot)\,f_j(\cdot), \tag{10}$$

where $V^{\circ}_{\max,j}$ is a reference capacity with units of flux. The other factors are dimensionless. The reversibility factor δ_j sets the reverse direction. The reference capacity $V^{\circ}_{\max,j}$ and saturation factor f_j describe enzyme capacity and substrate saturation. The enzyme-abundance ratio (e/e°) compares the current enzyme concentration e with the reference concentration e°, and the allosteric factor θ_j describes enzyme activity. Transcriptional and translational controls u_j and w_j act through the expression model and change the enzyme-abundance ratio (e/e°). They are distinct from the allosteric factor θ_j. When data are unavailable, the corresponding factor is set to one. Measurements or mechanistic models can then replace that default.

The reversibility factor δ_j has two values:

$$\delta_j \;=\; \begin{cases} 1, & \text{reaction } j \text{ reversible}, \\ 0, & \text{reaction } j \text{ irreversible}, \end{cases}. \tag{11}$$

When the reversibility factor $\delta_j = 1$, the lower bound is negative and reverse flux is allowed. When the reversibility factor $\delta_j = 0$, the lower bound is zero. A simple assignment compares the standard reaction free energy ΔG°_j with a reversibility threshold ΔG^{*}_j. This threshold rule is heuristic. A thermodynamic analysis should instead propagate stated activity ranges through the reaction free energy $\Delta G = \Delta G^{\circ} + RT_{\text{abs}} \ln Q$, where R is the gas constant, T_{abs} is absolute temperature, and Q is the reaction quotient, and then test whether the reaction free energy ΔG can change sign. eQuilibrator provides standard free-energy estimates for this calculation [16].

Enzyme kinetics set the forward capacity. The reference capacity is given by:

$$V^{\circ}_{\max,j} \;=\; k^{\circ}_{\text{cat},j}\, e^{\circ}, \tag{12}$$

where $k^{\circ}_{\text{cat},j}$ is the turnover number and e° is a reference enzyme concentration. BRENDA reports turnover numbers [17], and BioNumbers reports cellular abundance scales [18]. The saturation factor gives the fraction of the reference capacity $V^{\circ}_{\max,j}$ available at the current substrate concentration. For a single-substrate enzyme, the reaction mechanism is written as:

$$E + S_j \;\underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}}\; [ES] \;\xrightarrow{k_{\text{cat}}}\; E + P, \tag{13}$$

where E is free enzyme, S_j is substrate, $[ES]$ is the enzyme–substrate complex, and P is product. The rate constants k_1, k_{-1}, and k_{cat} describe substrate association, substrate dissociation, and catalysis, respectively. The quasi-steady-state approximation and enzyme conservation, $[E]_T = [E] + [ES]$, give $[ES] = [E]_T[S_j]/(K_{M,j} + [S_j])$, where $[E]_T$ is total enzyme concentration and $K_{M,j} = (k_{-1} + k_{\text{cat}})/k_1$ is the Michaelis constant. With the reaction rate $v_j = k_{\text{cat}}[ES]$ and maximum rate $V_{\max,j} = k_{\text{cat}}[E]_T$, the Michaelis–Menten law is given by [5]:

$$v_j \;=\; V_{\max,j}\,\frac{[S_j]}{K_{M,j} + [S_j]} \qquad \Longrightarrow \qquad f_j([S_j]) \;=\; \frac{v_j}{V_{\max,j}} \;=\; \frac{[S_j]}{K_{M,j} + [S_j]} \;\in [0,1], \tag{14}$$

where $[S_j]$ is the substrate concentration. The saturation factor f_j approaches zero at low substrate and one at saturation. Because the saturation factor f_j depends on the substrate concentration $[S_j]$, the bound becomes state-dependent.

Gene expression sets enzyme abundance. The balances for transcript m_j and protein p_j are given by:

$$\dot{m}_j = r_{X,j}\, u_j - (\theta_{m,j} + \mu)\, m_j + \lambda_j, \tag{15}$$
$$\dot{p}_j = r_{L,j}\, w_j\, m_j - (\theta_{p,j} + \mu)\, p_j, \tag{16}$$

where $r_{X,j}$ and $r_{L,j}$ are transcription and translation capacities, u_j is the dimensionless transcriptional control, w_j is the dimensionless translational control, $\theta_{m,j}$ and $\theta_{p,j}$ are degradation constants, and λ_j is the basal transcription rate. The growth rate μ adds to both loss rates. At steady state, the solutions are:

$$m_j^\star = \frac{r_{X,j}u_j + \lambda_j}{\theta_{m,j} + \mu}, \qquad p_j^\star = \frac{r_{L,j}w_j\, m_j^\star}{\theta_{p,j} + \mu}, \qquad (e/e^\circ) = \frac{p_j^\star}{e^\circ}. \tag{17}$$

Gene–protein–reaction rules map the steady-state protein abundance $p_j^\star$ to catalytic activity. These rules represent single enzymes, multisubunit complexes, and isozymes. Measured transcript or protein abundance can also set the enzyme-abundance ratio (e/e°) directly and produce condition-specific capacities.

Active and inactive configurations provide one statistical-mechanical description of allosteric, transcriptional, and translational control (Fig. 2). For an enzyme, promoter, or translation complex, the control is the statistical weight of the active set divided by the weight of all configurations. Let $\mathcal{S}_q$ be the configurations for process q, where q denotes catalysis, transcription, or translation. For a configuration $s \in \mathcal{S}_q$ with energy ϵ_s, the state weight, partition function, and state probability are given by:

$$\omega_s(\mathbf{x}) = g_s(\mathbf{x})e^{-\beta\epsilon_s}, \qquad Z_q(\mathbf{x}) = \sum_{r\in\mathcal{S}_q} \omega_r(\mathbf{x}), \qquad p_s(\mathbf{x}) = \frac{\omega_s(\mathbf{x})}{Z_q(\mathbf{x})}, \tag{18}$$

where $\beta = 1/(k_B T_{\text{abs}})$ is the inverse thermal energy, k_B is the Boltzmann constant, T_{abs} is absolute temperature, and Z_q is the partition function. The concentration factor $g_s(\mathbf{x})$ includes ligand concentration, binding stoichiometry, and degeneracy. For example, a state with binding stoichiometry n, effector concentration x, and concentration scale K has a concentration factor proportional to $(x/K)^n$.

Let $\mathcal{A}_q \subseteq \mathcal{S}_q$ contain the active configurations and let $\mathcal{I}_q = \mathcal{S}_q \setminus \mathcal{A}_q$ contain the inactive configurations. The control Φ_q is the probability of the active set:

$$\Phi_q(\mathbf{x}) = \sum_{s\in\mathcal{A}_q} p_s(\mathbf{x}) = \frac{\displaystyle\sum_{s\in\mathcal{A}_q} g_s(\mathbf{x})e^{-\beta\epsilon_s}}{\displaystyle\sum_{r\in\mathcal{S}_q} g_r(\mathbf{x})e^{-\beta\epsilon_r}} = \frac{Z_{\mathcal{A}_q}}{Z_{\mathcal{A}_q} + Z_{\mathcal{I}_q}}. \tag{19}$$

Here $Z_{\mathcal{C}} = \sum_{s\in\mathcal{C}} \omega_s$ is the partition sum over a configuration subset $\mathcal{C}$. Active states appear in the numerator and denominator; inactive states appear only in the denominator. Therefore $0 \leq \Phi_q \leq 1$ for the control Φ_q. An effector increases or decreases the control Φ_q according to the states it stabilizes. Choose configuration 0 as a reference, define the energy difference $\Delta\epsilon_s = \epsilon_s - \epsilon_0$, and denote the effective relative weight of state s by $K_{\star,s}$. The relative weights are then given by:

$$K_{\star,s} \equiv \frac{\omega_s/g_s}{\omega_0/g_0} = e^{-\beta\Delta\epsilon_s}, \qquad \frac{\omega_s}{\omega_0} = \frac{g_s(\mathbf{x})}{g_0(\mathbf{x})} K_{\star,s}. \tag{20}$$

With the reference weight ω_0 and reference concentration factor g_0 set to one, the state weight is $\omega_s = g_s(\mathbf{x})K_{\star,s}$. A term such as $K_2 f_{TL}$ combines the energetic factor $K_2 = e^{-\beta\Delta\epsilon_2}$ with the available ligand-bound transcription-factor fraction f_{TL}. Fitted state-weight parameters K_1 and K_2 may also include constant polymerase concentration, standard-state terms, or other fixed contributions. They need not be measured dissociation constants.

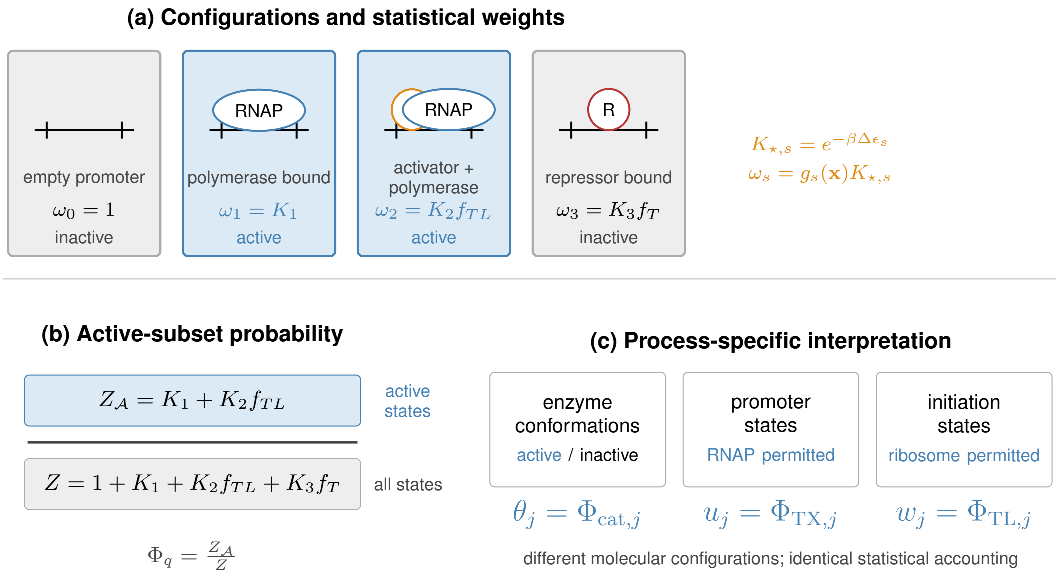

Figure 2: Active-subset construction of biological control factors. (a) Configuration weights combine relative energetics with regulator availability; blue states permit the process and gray states do not. (b) Active states enter the numerator, while all states enter the partition-function denominator. The promoter shown is schematic. (c) Specifying the relevant configurations gives allosteric activity θ_j, transcriptional control u_j, or translational control w_j from the same statistical accounting.

Moon, Voigt, and coworkers used this construction for inducible promoters [19]. The Lux promoter has state weights 1, K_1, and $K_2 f_{TL}$ for the empty, polymerase-bound, and polymerase–LuxR-bound states. Both polymerase-bound states permit transcription. The Tac promoter has state weights 1, K_1, and $K_2 f_T$ for the empty, polymerase-bound, and repressor-bound states. Only the polymerase-bound state permits transcription. Normalization gives the promoter controls:

$$u_{\mathrm{Lux}} = \frac{K_1 + K_2 f_{TL}}{1 + K_1 + K_2 f_{TL}}, \qquad u_{\mathrm{Tac}} = \frac{K_1}{1 + K_1 + K_2 f_T}. \tag{21}$$

Here the fraction f_{TL} is ligand-bound transcription factor and the fraction f_T is ligand-free transcription factor. Increasing inducer adds weight to an active Lux state and removes weight from an inactive Tac state. Both controls are special cases of Equation (19). A two-state model gives a familiar special case. If the inactive state has weight $\omega_I = 1$ and the effector-stabilized active state has weight $\omega_A = (x/K)^n e^{-\beta\Delta\epsilon}$, then the control is given by:

$$\Phi_q(x) = \frac{(x/K)^n e^{-\beta\Delta\epsilon}}{1 + (x/K)^n e^{-\beta\Delta\epsilon}}, \tag{22}$$

where the energy difference $\Delta\epsilon$ is the active-state energy relative to the inactive state. If the effector instead stabilizes the inactive state, its weight appears only in the denominator and the control decreases with the effector concentration x.

Equation (19) gives each process-specific factor:

$$\theta_j = \Phi_{\text{cat},j}, \qquad \bar{u}_j = \Phi_{\text{TX},j}, \qquad w_j = \Phi_{\text{TL},j}. \tag{23}$$

The allosteric factor θ_j multiplies catalytic capacity in Equation (10). The total transcriptional control $\bar{u}_j$ controls transcription, and the translational control w_j controls translation in Equation (16). These controls are distinct from the degradation constants $\theta_{m,j}$ and $\theta_{p,j}$. Split the transcriptionally active set into regulated and basal states. Let $\mathcal{A}_{\text{TX}}^{\text{reg}}$ contain the regulated states and $\mathcal{A}_{\text{TX}}^{\text{basal}}$ contain states that support transcription without the regulated input. The resulting split is given by:

$$\bar{u}_j = \frac{Z_{\mathcal{A}_{\text{TX}}^{\text{reg}}}}{Z_{\text{TX}}} + \frac{Z_{\mathcal{A}_{\text{TX}}^{\text{basal}}}}{Z_{\text{TX}}} = u_j + u_j^{\dagger}, \qquad \lambda_j \equiv r_{X,j} u_j^{\dagger}. \tag{24}$$

Here Z_{TX} is the total transcriptional partition function. The regulated transcriptional control u_j enters Equation (15). The basal control $u_j^{\dagger}$ defines the basal transcription rate λ_j. Moon, Voigt, and coworkers used this promoter-state construction [19]. Adhikari and coworkers extended it to transcriptional and translational control [20].

These factors require thermodynamic, kinetic, expression, and regulatory data. When data are unavailable, set the saturation factor $f_j \approx 1$, the enzyme-abundance ratio $(e/e^{\circ}) \approx 1$, and the allosteric factor $\theta_j \approx 1$. This gives the baseline bound:

$$-\delta_j V_{\max,j}^{\circ} \;\leq\; \hat{v}_j \;\leq\; V_{\max,j}^{\circ}. \tag{25}$$

Equation (25) retains only reversibility and reference capacity. Equation (10) shows which omitted factors require additional data. Each factor has a precedent in constraint-based modeling. Thermodynamics- based flux analysis uses free energies and metabolite activity ranges to exclude infeasible states [21]. Enzyme-constrained FBA uses turnover numbers and enzyme abundance; GECKO applies this approach at genome scale [22]. Metabolism-and-expression models couple transcription and translation to metabolism [23]. Regulatory FBA uses Boolean expression rules [24], whereas PROM uses data-derived interaction probabilities [25]. The allosteric factor θ_j represents direct regulation of enzyme activity. Equation (10) places these established terms in one notation. The balances above use aggregate transcription and translation capacities $r_{X,j}$ and $r_{L,j}$. A sequence-resolved model can instead calculate nucleotide, amino-acid, and energy demands from each gene and protein sequence and add those reactions to the stoichiometric matrix $\mathbf{S}$ [9]. Expression then becomes part of the flux calculation. Vilkhovoy and coworkers used this construction for cell-free protein synthesis [26]; the later capstone returns to that model.

5 Worked Example: Urea-Cycle Metabolism

This example uses thermodynamic and kinetic data to set flux bounds for a urea-cycle network with four cycle reactions, a competing nitric oxide synthase reaction, and exchange with the environment (left panel, Fig. 3). The inputs come from public, cross-organism databases rather than HL-60-specific measurements. The model uses the steady-state constraint $\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}$ from Equation (8) and the linear program in Equation (9). The application-specific quantities are the stoichiometric matrix $\mathbf{S}$, lower-bound vector $\boldsymbol{\ell}$, upper-bound vector $\mathbf{u}$, and objective coefficient vector $\mathbf{c}$. Argininosuccinate synthetase (v_1, EC 6.3.4.5) forms argininosuccinate. Argininosuccinate lyase (v_2, EC 4.3.2.1) produces arginine and fumarate. Arginase I (v_3, EC 3.5.3.1) produces ornithine and urea, and ornithine transcarbamylase (v_4, EC 2.1.3.3) regenerates citrulline. Nitric oxide synthase (v_5, EC 1.14.13.39) competes with arginase for arginine and produces nitric oxide and citrulline. The five enzymatic reactions involve eighteen metabolites. Fourteen metabolites cross the system boundary through exchange reactions b_1 through b_{14}. We use the secretion-positive convention $M_i \to \emptyset$ for boundary species M_i: positive flux denotes secretion, and negative flux denotes uptake. Arginine, citrulline, ornithine, and argininosuccinate remain inside the network. The resulting matrix has eighteen metabolite balances and nineteen reactions, so the stoichiometric matrix $\mathbf{S} \in \mathbb{R}^{18\times 19}$.

We first set the thermodynamic factor in Equation (11). eQuilibrator gave the following standard reaction free energies [16]: $\Delta G^{\circ}_{v_1} = -4.3$, $\Delta G^{\circ}_{v_2} = -5.5$, $\Delta G^{\circ}_{v_3} = -51.0$, $\Delta G^{\circ}_{v_4} = -30.3$, and $\Delta G^{\circ}_{v_5} = -1220$ kJ/mol. The reversibility threshold ΔG^{*}_{j} of Equation (11) was set to -10 kJ/mol. This is a heuristic cutoff, not a value derived from a stated physiological concentration range. The rule makes reactions v_1 and v_2 reversible and reactions v_3, v_4, and v_5 irreversible. Exchange reactions retain wide bounds in both directions because their direction depends on the medium. We next set the kinetic capacities in Equation (12). Turnover numbers came from BRENDA [17]. Each was combined with the reference enzyme abundance $e^{\circ} = 0.01$ mmol/gDW, an illustrative intracellular enzyme concentration consistent with BioNumbers [18]. Argininosuccinate synthetase and nitric oxide synthase have the turnover number $k^{\circ}_{\text{cat}} = 10.0\ \text{s}^{-1}$, which gives the reference capacities $V^{\circ}_{\text{max},v_1} = V^{\circ}_{\text{max},v_5} = 360\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$; arginase I has the turnover number $k^{\circ}_{\text{cat}} = 190.0\ \text{s}^{-1}$, giving the reference capacity $V^{\circ}_{\text{max},v_3} = 6840\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$; and ornithine transcarbamylase has the turnover number $k^{\circ}_{\text{cat}} = 410.0\ \text{s}^{-1}$, giving the reference capacity $V^{\circ}_{\text{max},v_4} = 14760\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$. Argininosuccinate lyase has the lowest turnover number, $k^{\circ}_{\text{cat},v_2} = 3.28\ \text{s}^{-1}$, which gives the reference capacity $V^{\circ}_{\text{max},v_2} = 118.08\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$, the smallest of the five capacities. Exchange reactions retain wide capacity bounds.

The model has no substrate, expression, or regulatory data, so setting the saturation factor f_j, enzyme-abundance ratio (e/e°), and allosteric factor θ_j to one yields the reported optimal flux distribution (right panel, Fig. 3). The enzymatic bounds therefore reduce to $-\delta_j V^{\circ}_{\text{max},j} \leq \hat{v}_j \leq V^{\circ}_{\text{max},j}$ for reactions v_1 through v_5. The fourteen exchange reactions use wide bounds of $-1000 \leq \hat{v}_j \leq 1000\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$. The objective is urea export. Because exchange reaction b_4 is written as urea $\to \emptyset$, we set its objective coefficient $c_{b_4} = 1$ and all other entries of the objective vector $\mathbf{c}$ to zero. The linear program [10] gave a unique optimum. Reactions v_1 through v_4 each carried $118.08\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$, the capacity of argininosuccinate lyase. The competing nitric oxide synthase branch v_5 carried zero flux because the objective assigns all available arginine to urea production. The optimal urea export was $118.08\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$, attained at the urea-export flux $\hat{v}_{b_4} = 118.08$, and every exchange flux carrying carbon or nitrogen into or out of the cycle, carbamoyl phosphate and aspartate on the uptake side, urea and fumarate on the secretion side, scaled to match it. Flux-variability analysis gave the same minimum and maximum for every reaction when the objective was held at its optimum. The optimal flux distribution is therefore unique to numerical tolerance.

The Monte Carlo analysis treats the database inputs as uncertain rather than exact and compares the effect of imputing the missing saturation factor (Fig. 4). We assigned lognormal uncertainty of about a factor of two to each turnover number and the reference abundance, and normal uncertainty of a few kilojoules per mole to each standard free energy. For four reactions, we also sampled substrate concentrations and Michaelis constants from Park and coworkers [27]. Argininosuccinate lyase lacked a measured substrate concentration, so its saturation factor remained at one. We then solved the linear program for ten thousand parameter draws using Algorithm 1. This Monte Carlo calculation produced a right-skewed urea-export distribution with median 105 and mean $156\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$ and a central ninety-five percent interval of 17 to 614. Varying only capacity and free-energy inputs gave a median of 113 $\mathrm{mmol\,gDW^{-1}\,h^{-1}}$. Adding the four measured saturation factors changed the median to 105, so those factors had little effect. Most of the uncertainty came from the sampled capacity of argininosuccinate lyase. The reversibility switches did not bind because the optimal cycle flux was forward, and the other four enzymes usually retained excess capacity. The nitric oxide synthase flux remained small overall. The missing saturation factor for argininosuccinate lyase matters more. When we imputed it from a physiological argininosuccinate concentration and a BRENDA Michaelis constant, the median fell to 50 $\mathrm{mmol\,gDW^{-1}\,h^{-1}}$ and the distribution widened toward zero. Better data for this limiting reaction would reduce the uncertainty most directly. This example used no HL-60-specific concentration, expression, or regulatory measurements. eQuilibrator set reaction direction, BRENDA set capacity scales, and the other factors remained at one. The result is limited mainly by argininosuccinate lyase reaction v_2, while the competing nitric oxide synthase reaction v_5 carries little flux. The next example adds allosteric and expression control to a reaction bound.

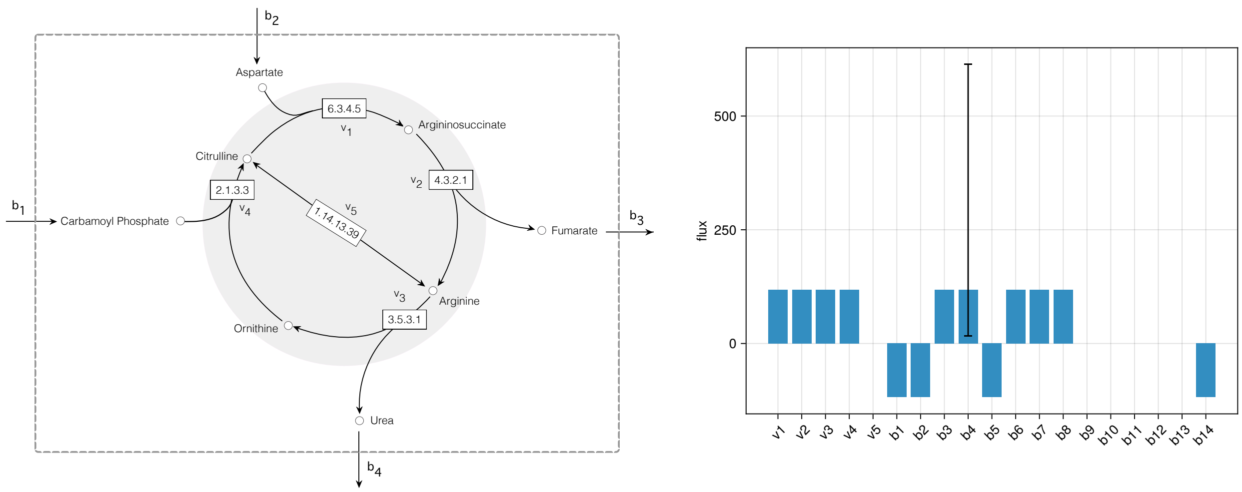

Figure 3: Urea-cycle network and optimal fluxes. Left: reactions v_1 through v_4 form the cycle, and v_5 is the competing nitric oxide synthase branch. Exchange reactions cross the dashed boundary; arginine, citrulline, ornithine, and argininosuccinate remain internal. Right: the optimal flux distribution for Equation (9). Positive exchange flux denotes secretion, and negative exchange flux denotes uptake. The cycle reactions carry 118.08 $\mathrm{mmol\,gDW^{-1}\,h^{-1}}$, the capacity of argininosuccinate lyase (v_2), while nitric oxide synthase reaction v_5 carries zero flux. The whisker on urea export b_4 shows the 2.5–97.5 percentile interval from the Monte Carlo calculation in Algorithm 1.

6 Worked Example: A Feedback-Inhibited Linear Pathway

The urea-cycle example used thermodynamic and kinetic factors but set the enzyme-abundance ratio (e/e°) and allosteric factor θ_j to one. This example includes both controls. The end product inhibits the existing enzyme and represses transcription of its gene. These mechanisms change enzyme activity and enzyme abundance, respectively, and they act on different timescales. We represent them in a linear pathway with one controlled step:

$$\emptyset \xrightarrow{r_0} X_1 \xrightarrow{r_1} X_2 \xrightarrow{r_2} X_3 \xrightarrow{r_3} \emptyset, \tag{26}$$

where r_0 is the committed uptake step, r_1 and r_2 are interconversions, and r_3 exports the end-product metabolite X_3. This end product controls both the activity and abundance of the enzyme

Algorithm 1 Monte Carlo propagation of urea-cycle parameter uncertainty

Require: nominal turnover numbers k°_{cat}, reference enzyme abundance e°, and standard reaction free energies ΔG°; nominal substrate concentrations $\overline{[S_j]}$ and Michaelis constants $\overline{K_{M,j}}$ where measured; log-space spread $\sigma_{\ln}$ and free-energy standard deviation $\sigma_{\Delta G}$; reversibility threshold ΔG^{*}; draw count N; random seed

seed the random generator
for draw $i = 1$ to N **do**
    sample enzyme abundance $e \leftarrow e^{\circ} \exp(\sigma_{\ln} Z_e)$ ▷ one shared abundance draw, $Z_e \sim \mathcal{N}(0,1)$
    for each enzymatic reaction j **do**
        $k_{\text{cat},j} \leftarrow k^{\circ}_{\text{cat},j} \exp(\sigma_{\ln} Z_{k,j}), \quad \Delta G_j \leftarrow \Delta G^{\circ}_j + \sigma_{\Delta G} Z_{\Delta G,j}$ ▷ independent $Z_{k,j}, Z_{\Delta G,j} \sim \mathcal{N}(0,1)$ per reaction
        set reversibility factor $\delta_j \leftarrow \mathbb{1}[\Delta G_j > \Delta G^{*}]$
        if substrate data measured for reaction j **then**
            $[S_j] \leftarrow \overline{[S_j]} \exp(\sigma_{\ln} Z_{S,j}), \quad K_{M,j} \leftarrow \overline{K_{M,j}} \exp(\sigma_{\ln} Z_{K,j})$ ▷ independent $Z_{S,j}, Z_{K,j} \sim \mathcal{N}(0,1)$
            set saturation factor $f_j \leftarrow [S_j]/(K_{M,j} + [S_j])$
        else
            set saturation factor $f_j \leftarrow 1$
        end if
        set capacity $V_{\max,j} \leftarrow 3600\, k_{\text{cat},j}\, e$; upper bound $b^{\text{U}}_j \leftarrow V_{\max,j} f_j$; lower bound $b^{\text{L}}_j \leftarrow -\delta_j V_{\max,j} f_j$
    end for
    $\hat{\mathbf{v}}^{(i)} \leftarrow \arg\max\{\mathbf{c}^{\top}\mathbf{v} : \mathbf{S}\mathbf{v} = \mathbf{0},\ \mathbf{b}^{\text{L}} \le \mathbf{v} \le \mathbf{b}^{\text{U}}\}$
end for
return per-reaction mean, standard deviation, and 2.5/50/97.5 percentiles of $\{\hat{\mathbf{v}}^{(i)}\}_{i=1}^{N}$

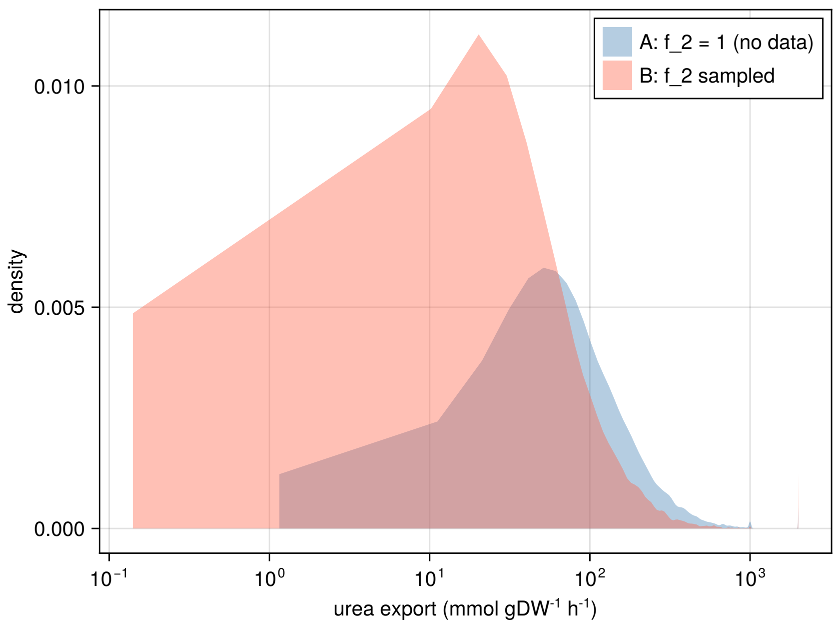

Figure 4: Sensitivity of urea export to saturation information. Configuration A (blue) samples four saturation factors derived from data reported by Park and coworkers [27] and holds the saturation factor $f_2 = 1$ at the argininosuccinate lyase bottleneck, whose substrate is unmeasured, giving a median urea export of 105 $\mathrm{mmol\,gDW^{-1}\,h^{-1}}$. Configuration B (red) imputes the saturation factor f_2 from a physiological argininosuccinate concentration and a BRENDA Michaelis constant and samples it as well, shifting the median to 50 $\mathrm{mmol\,gDW^{-1}\,h^{-1}}$ and widening the band toward zero. The horizontal axis is logarithmic.

for committed step r_0. We generated a kinetic reference with a Biochemical Systems Theory (BST) S-system [6, 7]. The model includes five dynamic states: metabolites X_1, X_2, and X_3, transcript m, and enzyme E_0. BSTModelKit.jl was used to integrate these states to steady state [8]. The end product enters the committed-step rate with kinetic order $-a$ and the transcription rate with kinetic order $-b$. Increasing the end-product concentration X_3 therefore lowers both enzyme activity and enzyme production. Metabolic fluxes obey the steady-state balance $\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}$, while transcript and protein balances include degradation and growth dilution. The committed-step rate v_{r_0} and BST activity factor θ_{BST} are given by:

$$\begin{aligned} v_{r_0} &= V^{\circ}_{\max,r_0}\,(e/e^{\circ})\,\theta_{\text{BST}}, \\ \theta_{\text{BST}} &= (X_3/X_3^{\circ})^{-a}, \qquad a = 0.4, \\ \text{transcription} &\propto (X_3/X_3^{\circ})^{-b}, \qquad b = 0.6, \end{aligned} \tag{27}$$

where the reference end-product concentration $X_3^{\circ} = 1$ in the concentration units used by the simulation. The normalized ratios are dimensionless.

The kinetic reference combines expression timescales with metabolic feedback and gives a steady-state pathway throughput $T^{\star} \approx 2.66$ (Fig. 5). It uses a 330-residue enzyme, a 990-nucleotide gene, polymerase elongation at 60 nucleotides per second, and ribosome elongation at 16 amino acids per second. These values provide illustrative expression timescales. The steady-state abundance also depends on an mRNA half-life of 2.5 minutes, a protein half-life of 35 minutes, and a culture doubling time of 60 minutes. The same growth rate μ appears in both balances, but its relative contribution differs because transcript and protein have different degradation rates. With transcript degradation rate θ_m and protein degradation rate θ_p, growth accounts for about four percent of the total transcript clearance rate $\theta_m + \mu \approx 0.289$ min^{-1} and about thirty-seven percent of the total enzyme clearance rate $\theta_p + \mu \approx 0.031$ min^{-1}. At steady state, every reaction in the linear chain carries the same pathway throughput T. Because export is first order in the end-product concentration X_3, the concentration is $X_3 = T/k_3$, where k_3 is the first-order export constant. Substitution of the activity and expression controls gives one consistency condition for the end-product concentration X_3. The numerical solution was the steady-state concentration $X_3^{\star} \approx 4$, with activity factor $\theta^{\star}_{\text{BST}} \approx 0.574$, an enzyme-abundance ratio $(e/e^{\circ})^{\star} \approx 0.464$. The basal transcription rate λ, defined by Equations (15) and (24), keeps expression above zero. The kinetic orders $a = 0.4$ and $b = 0.6$ make both controls affect the result; neither control alone produces the reference throughput.

The FBA comparison keeps the chain's stoichiometry unchanged and places the entire dual-control content on the committed-step bound (Fig. 5). The bound on committed-step flux $\hat{v}_{r_0}$ is given by:

$$0 \;\leq\; \hat{v}_{r_0} \;\leq\; V^{\circ}_{\max,r_0}\,(e/e^{\circ})\,\theta_{\text{FBA}}(X_3), \qquad V^{\circ}_{\max,r_0} = 10, \tag{28}$$

which is the expression-and-regulation form of Equation (10). The thermodynamic and saturation factors remain at one. To avoid copying the BST rate law into the FBA bound, we represent the FBA activity factor θ_{FBA} with a bounded two-state Hill function using half-saturation scale $K = 5$ and Hill coefficient $n = 2$:

$$\theta_{\text{FBA}}(X_3) \;=\; \frac{K^n}{K^n + X_3^n}, \qquad K = 5, \quad n = 2, \tag{29}$$

the repressive form of Equation (22). The active state has unit weight, and the inactive state has weight $(X_3/K)^n$. The half-saturation scale K and Hill coefficient n were specified independently of the BST kinetic order a. Both control factors were evaluated from the steady-state enzyme abundance and end-product concentration X_3, not from the reference flux. Four FBA solutions used the same stoichiometric matrix and differed only in the bound on committed step r_0. With both the expression and activity factors set to one, the predicted flux was 10. Expression control alone reduced it to 4.64, activity control alone reduced it to 6.10, and both controls reduced it to 2.83. The dual-control result is within about six percent of the BST reference value, 2.66.

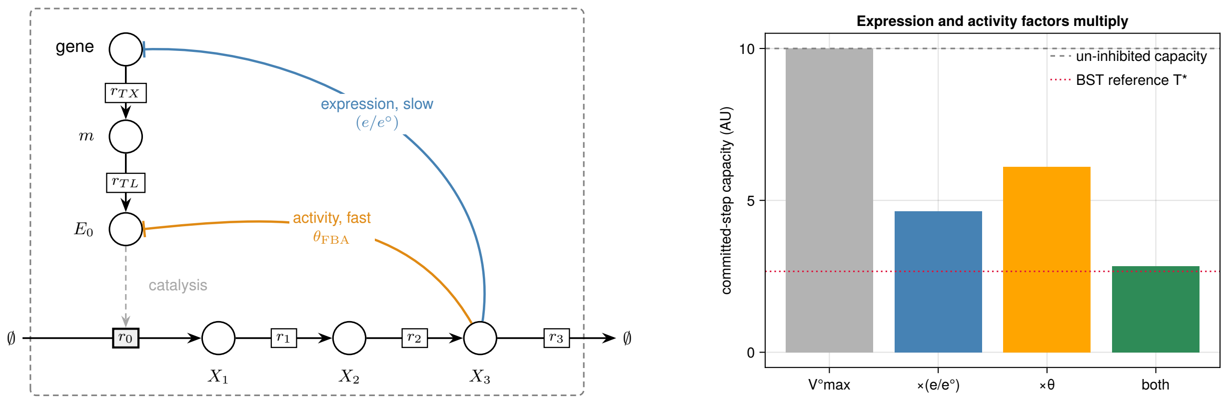

Figure 5: Dual-control feedback example. Left: the pathway and the transcription and translation steps that produce enzyme E_0. The end product X_3 represses enzyme expression and inhibits enzyme activity. Right: predicted committed-step flux r_0 with neither control, each control alone, and both controls. Expression control gives 4.64, activity control gives 6.10, and both give 2.83. The BST reference is pathway throughput $T^{\star} \approx 2.66$, and the dashed line marks the uninhibited capacity of 10.

The parameter sweep addresses uncertainty in the Hill parameters, which were specified as half-saturation scale $K = 5$ and Hill coefficient $n = 2$ rather than fitted (Fig. 6). We therefore varied the scale K from 3 to 8 and coefficient n from 1 to 3, recomputed the FBA activity factor θ_{FBA} at the reference end-product concentration X_3, and solved the program at each grid point. With both controls, the predicted flux ranged from about 1.4 to 4.1 and included the BST value of 2.66. The nominal parameters gave 2.83. Activity control alone kept the flux above about 2.97, while expression control alone kept it at 4.64. Thus, neither single-control model included the BST value over the tested parameter range, but the dual-control model did.

7 Integration in Practice

The worked examples used small networks whose reaction count n and metabolite count m differed by one, so $n - m = 1$. This section considers two larger applications. The first combines metabolism, expression, and regulation in a dynamic cell-free model. The second changes reaction bounds to design a glycan-producing strain. Vilkhovoy and coworkers built a sequence-specific model of *E. coli* cell-free protein synthesis [26]. Transcription and translation reactions derived from DNA and mRNA sequences appear in the stoichiometric matrix with metabolism. Promoter and ribosome control functions u_j and w_j follow the biophysical models of Moon, Voigt, Adhikari, and coworkers [19, 20]. Dai, Horvath, and Varner extended the sequence-specific formulation to a dynamic constraint-based model that solved an optimization problem over short time intervals while enforcing time-dependent metabolite constraints [12]. A later integrated model updated thermodynamic, kinetic, expression, and regulatory bounds during the reaction and was compared

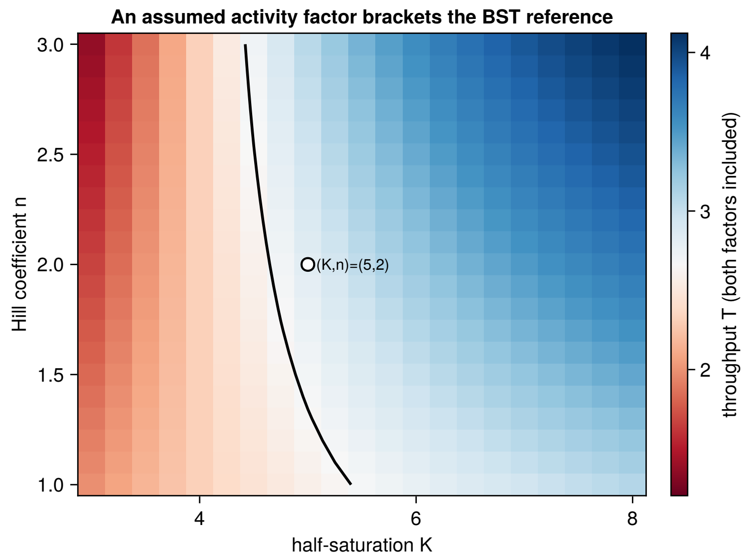

Figure 6: Sensitivity to the assumed activity-factor shape. The plot shows the dual-control flux while the half-saturation scale K and Hill coefficient n vary in the FBA activity factor $\theta_{\mathrm{FBA}} = K^n/(K^n + X_3^n)$. The color scale is centered on the BST reference $T^\star \approx 2.66$, and the black contour marks equal FBA and BST fluxes. The predicted flux ranges from about 1.4 to 4.1. The marked nominal parameter pair, $(K, n) = (5, 2)$, gives 2.83.

with time courses for 63 metabolites, mRNA, protein, and enzyme activity [13]. Cell-free batches have no growth dilution, so the growth rate $\mu = 0$, but their molecular pools still change. On a fixed concentration basis, their balance is therefore the dynamic form $\mathbf{Sv} = d\mathbf{x}/dt$ for cell-free flux vector $\mathbf{v}$, not the steady-state form $\mathbf{Sv} = \mathbf{0}$.

Wayman and coworkers used a constraint-based model for strain design. They added the N-linked glycosylation pathway of *Campylobacter jejuni* onto a genome-scale reconstruction of *E. coli* metabolism and searched for gene deletions that couple designer-glycan production to growth [14]. Each candidate knockout sets a reaction bound to zero, after which the linear program is solved again. Enzyme attenuation or overexpression can instead decrease or increase the enzyme-abundance ratio (e/e°) in Equation (10). The single Δgnd deletion, drawn from the model-identified strain families, increased measured glycan production nearly threefold relative to wild type. The model identified candidate perturbations; it did not predict the magnitude of this experimental increase. The two applications use bounds for different purposes: the cell-free model uses measurements to estimate fluxes, whereas the glycan model changes bounds to test possible interventions. Strain design also requires a discrete search over deletions or expression changes, which can be formulated as a bilevel program [15].

8 Outlook

The worked examples used standard free energies, turnover numbers, and regulatory models to set the factors in Equation (10). Time-course data can instead be used to estimate turnover numbers, saturation constants, allosteric factors θ_j, and expression controls u_j and w_j. Tabulated values remain useful as prior information when dynamic data are unavailable. The same approach can be applied reaction by reaction in genome-scale models. These models have many flux distributions that satisfy the steady-state balance $\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}$, so condition-specific bounds are especially important. Setting an unknown factor to one preserves feasibility but may leave many fluxes unresolved. Time-course data also allow models to retain accumulation and growth dilution. Cell-free systems have no growth dilution, but their molecular pools can still accumulate, whereas growing cultures can exhibit both effects. When measurements resolve these dynamics, the dynamic balance in Equation (6) provides the appropriate flux constraint. For a fixed-basis cell-free batch, it reduces to the accumulation balance $\mathbf{S}\mathbf{v} = d\mathbf{x}/dt$ for cell-free flux vector $\mathbf{v}$ and concentration vector $\mathbf{x}$, the form used in our earlier dynamic constraint-based model [12]. The steady-state approximation should be used only when accumulation and growth dilution are negligible.

Code and data availability

Source code, generated data, and instructions for reproducing the worked examples and figures are available at `https://github.com/varnerlab/MRW-BTC4-Chapter-Varner`. The repository includes the Julia project and manifest files needed to reproduce the computational environment.

References

[1] Aarash Bordbar, Jonathan M. Monk, Zachary A. King, and Bernhard O. Palsson. Constraint-based models predict metabolic and associated cellular functions. *Nature Reviews Genetics*, 15(2):107–120, 2014. doi: 10.1038/nrg3643.

[2] A. A. Esener, J. A. Roels, and N. W. F. Kossen. Theory and applications of unstructured growth models: Kinetic and energetic aspects. *Biotechnology and Bioengineering*, 25(12): 2803–2841, 1983. doi: 10.1002/bit.260251202.

[3] Michael M. Domach, S. K. Leung, R. E. Cahn, G. G. Cocks, and Michael L. Shuler. Computer model for glucose-limited growth of a single cell of Escherichia coli B/r-A. *Biotechnology and Bioengineering*, 26(3):203–216, 1984. doi: 10.1002/bit.260260303.

[4] P. Wu, N. G. Ray, and Michael L. Shuler. A single-cell model for CHO cells. *Annals of the New York Academy of Sciences*, 665:152–187, 1992. doi: 10.1111/j.1749-6632.1992.tb42583.x.

[5] Leonor Michaelis and Maud L. Menten. Die kinetik der invertinwirkung. *Biochemische Zeitschrift*, 49:333–369, 1913.

[6] Michael A. Savageau. *Biochemical Systems Analysis: A Study of Function and Design in Molecular Biology.* Addison-Wesley, Reading, MA, 1976.

[7] Michael A. Savageau, Eberhard O. Voit, and Douglas H. Irvine. Biochemical systems theory and metabolic control theory: 1. Fundamental similarities and differences. *Mathematical Biosciences*, 86(2):127–145, 1987. doi: 10.1016/0025-5564(87)90007-1.

[8] Sandra Vadhin and Jeffrey D. Varner. BSTModelKit.jl: A Julia package for constructing, solving, and analyzing biochemical systems theory models, 2026.

[9] Timothy E. Allen and Bernhard Ø. Palsson. Sequence-based analysis of metabolic demands for protein synthesis in prokaryotes. *Journal of Theoretical Biology*, 220(1):1–18, 2003.

[10] Jeffrey D. Orth, Ines Thiele, and Bernhard Ø. Palsson. What is flux balance analysis? *Nature Biotechnology*, 28(3):245–248, 2010. doi: 10.1038/nbt.1614.

[11] Laurent Heirendt, Sylvain Arreckx, Thomas Pfau, Sebastian N. Mendoza, Anne Richelle, Almut Heinken, Hulda S. Haraldsdóttir, Jacek Wachowiak, Sarah M. Keating, Vanja Vlasov, Sigrid Magnusdóttir, Chiam Yu Ng, German Preciat, Alise Žagare, Siu H. J. Chan, Maike K. Aurich, Catherine M. Clancy, Jennifer Modamio, John T. Earls, David S. Wishart, and Ronan M. T. Fleming. Creation and analysis of biochemical constraint-based models: the COBRA toolbox v3.0. *Nature Protocols*, 14(3):639–702, 2019. doi: 10.1038/s41596-018-0098-2.

[12] David Dai, Nicholas Horvath, and Jeffrey D. Varner. Dynamic sequence specific constraint-based modeling of cell-free protein synthesis. *Processes*, 6(8):132, 2018. doi: 10.3390/pr6080132.

[13] M. Vilkhovoy, S. Dammalapati, S. Vadhin, A. Adhikari, and J. Varner. Integrated constraint-based modeling of E. coli cell-free protein synthesis. *bioRxiv*, 2023. doi: 10.1101/2023.02.10.528035. preprint.

[14] Joseph A. Wayman, Cameron Glasscock, Thomas J. Mansell, Matthew P. DeLisa, and Jeffrey D. Varner. Improving designer glycan production in Escherichia coli through model-guided metabolic engineering. *Metabolic Engineering Communications*, 9:e00088, 2019. doi: 10.1016/j.mec.2019.e00088.

[15] Anthony P. Burgard, Priti Pharkya, and Costas D. Maranas. OptKnock: A bilevel programming framework for identifying gene knockout strategies for microbial strain optimization. *Biotechnology and Bioengineering*, 84(6):647–657, 2003. doi: 10.1002/bit.10803.

[16] Moritz E. Beber, Mattia G. Gollub, Dana Mozaffari, Kevin M. Shebek, Avi I. Flamholz, Ron Milo, and Elad Noor. eQuilibrator 3.0: a database solution for thermodynamic constant estimation. *Nucleic Acids Research*, 50(D1):D603–D609, 2022. doi: 10.1093/nar/gkab1106.

[17] Antje Chang, Lisa Jeske, Sandra Ulbrich, Julia Hofmann, Julia Koblitz, Ida Schomburg, Meina Neumann-Schaal, Dieter Jahn, and Dietmar Schomburg. BRENDA, the ELIXIR core data resource in 2021: new developments and updates. *Nucleic Acids Research*, 49(D1):D498–D508, 2021. doi: 10.1093/nar/gkaa1025.

[18] Ron Milo, Paul Jorgensen, Uri Moran, Griffin Weber, and Michael Springer. BioNumbers—the database of key numbers in molecular and cell biology. *Nucleic Acids Research*, 38:D750–D753, 2010. doi: 10.1093/nar/gkp889.

[19] Tae Seok Moon, Chunbo Lou, Alvin Tamsir, Brynne C. Stanton, and Christopher A. Voigt. Genetic programs constructed from layered logic gates in single cells. *Nature*, 491(7423):249–253, 2012. doi: 10.1038/nature11516.

[20] Abhinav Adhikari, Michael Vilkhovoy, Sandra Vadhin, Ha Eun Lim, and Jeffrey D. Varner. Effective biophysical modeling of cell free transcription and translation processes. *Frontiers in Bioengineering and Biotechnology*, 8:539081, 2020. doi: 10.3389/fbioe.2020.539081.

[21] Christopher S. Henry, Linda J. Broadbelt, and Vassily Hatzimanikatis. Thermodynamics-based metabolic flux analysis. *Biophysical Journal*, 92(5):1792–1805, 2007. doi: 10.1529/biophysj.106.093138.

[22] Benjamín J. Sánchez, Cheng Zhang, Avlant Nilsson, Petri-Jaan Lahtvee, Eduard J. Kerkhoven, and Jens Nielsen. Improving the phenotype predictions of a yeast genome-scale metabolic model by incorporating enzymatic constraints. *Molecular Systems Biology*, 13(8):935, 2017. doi: 10.15252/msb.20167411.

[23] Edward J. O'Brien, Joshua A. Lerman, Roger L. Chang, Daniel R. Hyduke, and Bernhard Ø. Palsson. Genome-scale models of metabolism and gene expression extend and refine growth phenotype prediction. *Molecular Systems Biology*, 9:693, 2013. doi: 10.1038/msb.2013.52.

[24] Markus W. Covert, Christophe H. Schilling, and Bernhard Palsson. Regulation of gene expression in flux balance models of metabolism. *Journal of Theoretical Biology*, 213(1):73–88, 2001. doi: 10.1006/jtbi.2001.2405.

[25] Sriram Chandrasekaran and Nathan D. Price. Probabilistic integrative modeling of genome-scale metabolic and regulatory networks in Escherichia coli and Mycobacterium tuberculosis. *Proceedings of the National Academy of Sciences*, 107(41):17845–17850, 2010. doi: 10.1073/pnas.1005139107.

[26] Michael Vilkhovoy, Nicholas Horvath, Che-Hsiao Shih, Joseph A. Wayman, Kara Calhoun, James Swartz, and Jeffrey D. Varner. Sequence specific modeling of E. coli cell-free protein synthesis. *ACS Synthetic Biology*, 7(8):1844–1857, 2018. doi: 10.1021/acssynbio.7b00465.

[27] Junyoung O. Park, Joshua D. Rabinowitz, et al. Metabolite concentrations, fluxes and free energies imply efficient enzyme usage. *Nature Chemical Biology*, 12(7):482–489, 2016. doi: 10.1038/nchembio.2077.